

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1)

MA1513

Lecture Problem Set 4

Instructions

- To be discussed during lecture in week 4 (Aug 31 and Sep 1, 2026).
 - Note that you do not need to submit the answers to this problem set.
 - To prepare for the problem set, students should view the "Week 4 key video" before the lecture.
 - Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.
1. For each of the following linear transformations, find the standard matrix by two approaches: (i) finding the images of the standard basis vectors; (ii) stacking method.

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that

$$T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad \text{and} \quad T\left(\begin{pmatrix} 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}.$$

(b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that

$$T\left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad T\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad T\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

2. For each of the following matrices,

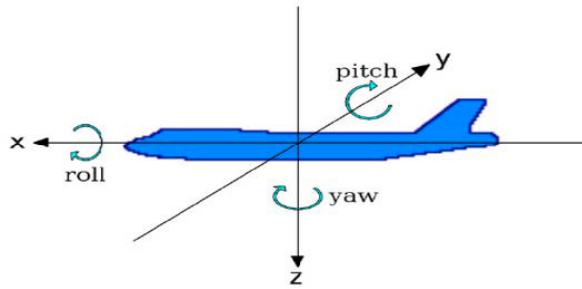
$$(a) \mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}, \quad (b) \mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad (c) \mathbf{C} = \begin{pmatrix} -1 & 1 & 1 \\ -2 & 2 & 1 \\ 2 & -1 & 0 \end{pmatrix}.$$

- (i) find all the eigenvalues of the matrix;
 - (ii) use the answer in (i) to determine whether the matrix is non-singular.
 - (iii) find a basis for the eigenspace associated with each eigenvalue of the matrix;
 - (iv) determine whether the matrix is diagonalizable;
 - (v) find a non-singular matrix $\mathbf{P}$ that diagonalize the matrix if it is diagonalizable.
3. A certain species of animals has two life stages: juvenile (up to 1 year old) and adult. Each year,

- an adult produces an average of 1.5 juveniles;
- 40% of the juveniles survive to become adults;
- 70% of the adults survive.

Let $\mathbf{x}_n = \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .

- Find the stage matrix $\mathbf{A}$ such that $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$.
 - Suppose the number of juveniles and adults are 12500 and 10000 respectively in year 2020. Find j_{2021} , a_{2021} , j_{2022} , a_{2022} .
 - Explain why the proportion of juveniles to adults remain the same over the years 2020, 2021 and 2022. Do you expect the proportion to stay constant in the long run?
4. (MATLAB) The terms yaw, pitch and roll are used in aerospace industry to describe the maneuvering of an aircraft.



Referring to the diagram above, a yaw is a rotation in the xy -plane. A yaw by an angle θ (clockwise) is a linear transformation with standard matrix

$$Y = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A pitch is a rotation in the xz -plane. A pitch by an angle α (clockwise) is a linear transformation with standard matrix

$$P = \begin{pmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{pmatrix}.$$

A roll is a rotation in the yz -plane. A roll by an angle β (clockwise) is a linear transformation with standard matrix

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{pmatrix}.$$

If we perform a yaw by an angle θ followed by a pitch by an angle α , the composite transformation is given by the product YP .

Similarly, a yaw with angle θ , followed by a pitch with angle α , and then a roll with angle β is given by the product YPR . The effect of such a combination of rotations is again a rotation about a certain axis. This axis is given by an eigenvector of the transformation matrix that corresponds to the eigenvalue 1 (why?).

Suppose an aircraft perform a yaw with 20° , followed by a pitch with 30° , and then a roll with 15° .

- (a) Write down the matrix $\mathbf{Q}$ of the composite transformation.
- (b) Check that 1 is an eigenvalue of $\mathbf{Q}$.
- (c) Find the axis of rotation of the composite transformation.

Hints and selected answers

1. (a) $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (only need to use two of the three equalities to find the standard matrix);
(b) $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ (need to use all the three equalities to find the standard matrix).

2. (a) eigenvalues 0 and 2; $\{(-1, 1, 0)^T\}$ basis for E_0 and $\{(1, 1, 0)^T\}$ basis for E_2 .
(b) eigenvalue 1; $\{(0, 0, 1)^T\}$ basis for E_1 .
(c) eigenvalues -1 and 1 ; $\{(-1, -1, 1)^T\}$ basis for E_{-1} and $\{(1, 2, 0)^T, (1, 0, 2)^T\}$ basis for E_1 .

We determine whether a 3×3 matrix is diagonalizable by checking whether it has three linearly independent eigenvectors. The maximum number of linearly independent eigenvectors can be obtained from the basis vectors for all the eigenspaces.

3. (a) $\begin{pmatrix} 0 & 1.5 \\ 0.4 & 0.7 \end{pmatrix}$.
(b) $j_{2021} = 15000, a_{2021} = 12000, j_{2022} = 18000, a_{2022} = 14400$.
(c) Find the eigenvectors of the stage matrix.
4. (a) Input the three matrices Y, P and R. The MATLAB command to input sine and cosine functions are $\sin(x)$, $\cos(x)$ (where x is some angle in radian).
For example, to input $\cos(20^\circ)$, type: `>> cos(pi*20/180)`.
After entering Y, P and R, use MATLAB to compute the matrix multiplication
`>> Q=Y*P*R`.
(b) Use the MATLAB command `>> [A B] = eig(Q)`.
The diagonal entries of B give the eigenvalues of Q. The three columns of A give the corresponding eigenvectors.
(c) The third column of A is the vector that gives the axis of rotation for the composite transformation.